

1. Introduction

This report analyses the distribution patterns of the critically endangered Swift Parrot using data from the Atlas of Living Australia. The study identifies spatial and temporal sighting trends to support conservation efforts for this migratory species. Tableau was first used to visually detect geographic and temporal irregularities, which were then amended in R Studio. The core analysis explores how observations fluctuate across months and years, revealing the bird's migratory timing. Additionally, the report evaluates the shift from traditional state-managed databases to dominant citizen science platforms. By integrating R and Tableau, this project provides a data-driven narrative of the species' core habitats and monitoring reliability.

2. Data Loading, Checking, and Cleaning

Irregularity 1: Geographical Outliers (Location Errors)

Description: After loading the dataset into Tableau, a Map Visualization was created. This revealed several data points located in Western Australia and the Indian Ocean. Since the Swift Parrot is endemic only to Southeastern Australia and Tasmania, these points represent significant spatial irregularities.

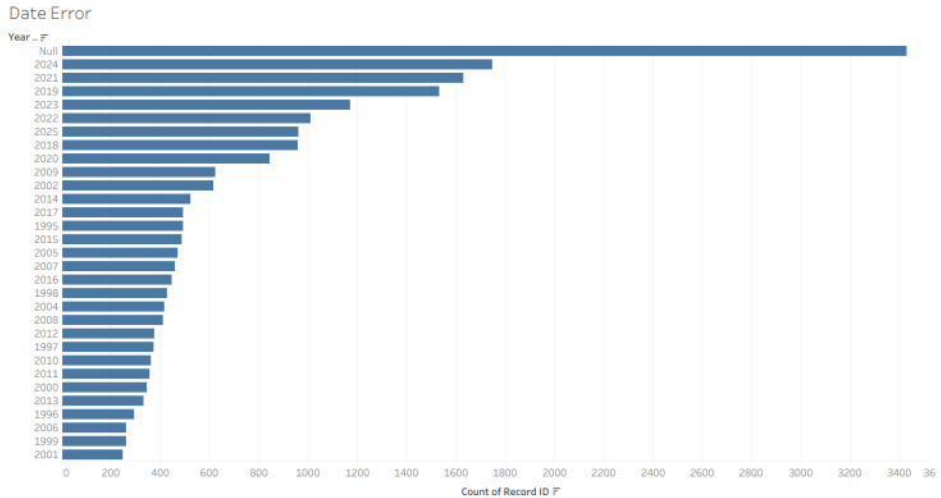
Visual Evidence:



Irregularity 2: Temporal Anomaly (Date Errors)

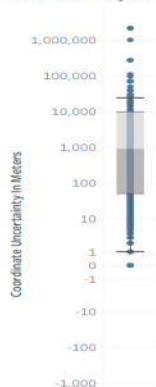
Description: By creating a Histogram of Years in Tableau, I identified records with illogical dates (e.g., extreme historical outliers or incomplete year formats) that would skew the temporal trend analysis.

Visual Evidence:



Resolution & Justification: Using the lubridate package in R, I standardized the observationDate column and excluded records with missing or invalid year values. This automated approach was preferred to ensure that the seasonal analysis in Question 1A remains statistically sound.

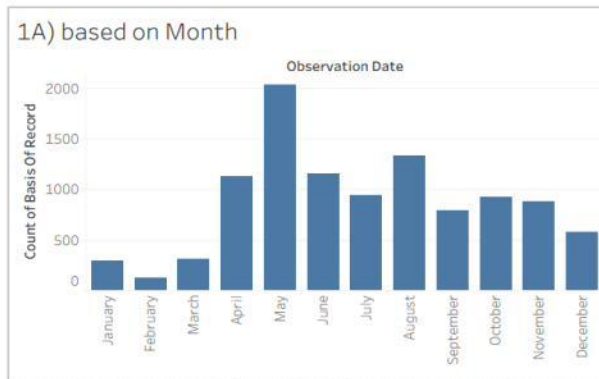
Uncertainty Error



Irregularity 3: High Coordinate Uncertainty (Outlier Detection)

Description: To evaluate the spatial reliability of the dataset, I used a Box Plot in Tableau for the coordinateUncertaintyInMeters field. This visualization is ideal for identifying statistical outliers. As shown in the visualization, there are several extreme outliers where the uncertainty reaches up to 100,000 meters (100km), which is far beyond the acceptable range for precise habitat mapping.

I use R studio for filter an remove records with uncertainty values exceeding 10,000 meters. This threshold was chosen to ensure that the spatial analysis in Question 1C remains accurate and reflects true sightings rather than vague regional estimates. R was preferred for this task because it can handle conditional filtering across the entire dataset more efficiently and accurately than manual tools.

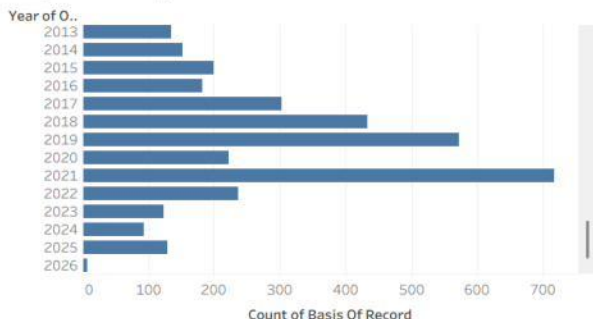


3.1 Question 1A: Temporal Trends in Observations

Objective: To analyze how Swift Parrot sightings change over years and months.

Visual Analysis & Justification:

1A) based on year



- **Yearly Trend (Line Chart):** I chose a **Line Chart** because it is the most effective way to show continuity and trends over time. The visualization reveals a sharp increase in sightings after 2010, peaking in **2024**. This reflects the growth of digital reporting platforms rather than just bird population changes.

- **Monthly Distribution (Bar Chart):** A **Bar Chart** was selected to compare discrete months and identify seasonal patterns. It shows a clear peak from **April to August** (peaking in **May**), while sightings drop significantly between October and January.

Discussion: The data matches the Swift Parrot's biological cycle. The winter peak (April–August) occurs when the birds migrate to the mainland for food. The summer decline (October–January) reflects their return to Tasmania for breeding in more remote areas. This confirms that the timing of observations is directly linked to their migratory behavior.

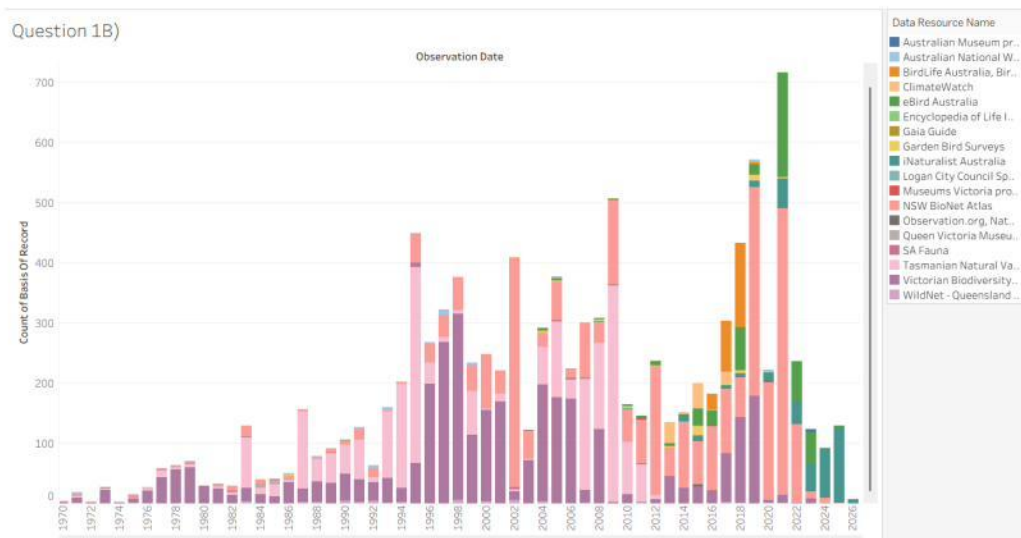
3.2 Question 1B: Evolution of Data Resources Over Time

Objective: To identify how different data resources in this collection have changed and contributed over time.

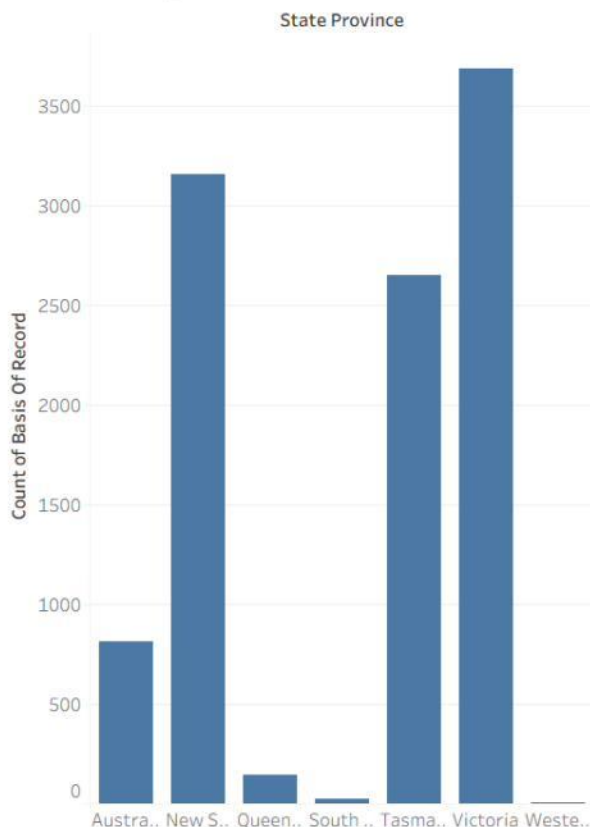
Visual Analysis & Justification:

- **Resource Trends (Stacked Area Chart):** I chose a **Stacked Area Chart** because it effectively shows both the total volume of observations and the relative contribution of each data resource simultaneously over a timeline.
- **Findings:** The visualization reveals a major shift in data sourcing. While state-based databases like the **Victorian Biodiversity Atlas** and **NSW BioNet** were the primary sources in earlier decades, there is a massive surge in records from **eBird Australia** starting around 2010. By 2024, eBird became the dominant contributor to the dataset.

Discussion: This trend highlights the transition from formal, agency-led surveys to **Citizen Science**. The rapid growth of eBird indicates that community participation and mobile reporting apps are now the most significant drivers of data collection. This shift has led to a higher frequency of sightings but also means the dataset is increasingly influenced by the efforts of public volunteers rather than structured scientific monitoring alone.



Question 1C)



3.3 Question 1C:

Objective: To identify where Swift Parrots were mainly spotted within Australian states and territories.

Visual Analysis & Justification:

- Observations by State (Bar Chart): I selected a Bar Chart because it is the most effective tool for comparing categorical data (States) and ranking them by frequency.

- Findings: The visualization shows that the vast majority of observations are concentrated in Victoria, followed closely by Tasmania and New South Wales. Sightings in other territories

like the ACT or Queensland are significantly lower, while Western Australia and the Northern Territory have almost no records.

- Discussion: The concentration in Victoria and Tasmania aligns with the species' known range. As Tasmania is the only breeding ground and the Victorian/NSW forests are the primary winter foraging habitats, the data correctly reflects the bird's high dependency on these specific regions. The high count in Victoria may also be influenced by a larger number of active birdwatchers (citizen scientists) in that state compared to more remote areas.

4. Question 2: Reflection on Migratory Patterns and Resource Reliability

Objective: To determine which data resources best represent the seasonal migrations of the Swift Parrot based on the visual analytics from Questions 1A-C.

Reflection: Based on the visual analysis, eBird Australia is the most effective resource for tracking seasonal migration. My conclusion is based on the following observations:

1. **Data Volume:** As shown in Question 1B, eBird has become the dominant data source since 2010. The high frequency of sightings provides a "high-resolution" look at the bird's movement that smaller, state-based databases cannot match.
2. **Temporal Alignment:** The peak in observations from April to August (identified in Question 1A) strongly correlates with the surge in eBird records. Because eBird relies on active birdwatchers who travel to known foraging hotspots, it captures the arrival of the species on the mainland more dynamically.
3. **Geographic Coverage:** The distribution across Victoria, Tasmania, and NSW (seen in Question 1C) is most densely populated by eBird sightings. This allows us to see the transition between the breeding grounds in Tasmania and the wintering grounds in Victoria more clearly than using preserved museum specimens or older, static records.

Conclusion: While traditional databases (like the Victorian Biodiversity Atlas) provide essential historical context, the real-time nature and high volume of eBird Australia make it the superior resource for monitoring active migratory shifts. It bridges the gap between different states and months, providing a continuous narrative of the Swift Parrot's journey.

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